# Yunkyeong Kang

M.S. Candidate, Graduate School of Engineering Biology, KAIST

Graduate Researcher, Synthetic Biology and Bioengineering Research Center, KRIBB

Daejeon, Republic of Korea · riveryk@g.skku.edu · tinykrex.github.io · github.com/TinyKRex · LinkedIn

**I track down errors and cut out wasted steps — in GMP quality assurance, and in the research workflows that generate the evidence.**

## PROFILE

1 year 8 months in GMP quality assurance at a biologics plant: taking the English record at MFDS and FDA inspections, answering the supply-chain lines of questioning, investigating deviations and managing change control. Now in graduate research on the other end of the same problem — finding new enzymes in metagenomes with protein language model embeddings and cell-free, biofoundry-scale validation, building both the pipelines that select and the assays that test. Looking for positions in quality and regulatory affairs, including RA documentation and R&D planning, as well as in AI-driven bio R&D. Business-level English (OPIc IH). Available from January 2027.

## RESEARCH INTERESTS

Protein language model embeddings · Metagenomic enzyme discovery · Cell-free protein synthesis and biofoundry automation · Design–Build–Test–Learn and active-learning loops · Structure-based interpretation of function · GMP quality systems and regulated evidence

## EDUCATION

### M.S. in Engineering Biology — KAIST, Graduate School of Engineering Biology

Feb 2025 – Feb 2027 (expected) · Advisor: Prof. Haseong Kim · KAIST–KRIBB joint program

Thesis: *Improving the Efficiency of Metagenomic Enzyme Screening with an Embedding-Based Iterative Learning Strategy*

### B.S. in Biological Science — Sungkyunkwan University

Mar 2016 – Feb 2023 · Magna Cum Laude · major GPA 4.31 / 4.5

Thesis: *Plant Breeding Techniques for Crop Improvement: Focusing on Genome Editing*

## RESEARCH EXPERIENCE

### Graduate Researcher — Synthetic Biology and Bioengineering Research Center, KRIBB

Feb 2025 – present, Daejeon · KAIST–KRIBB joint graduate program

Building an iterative Design–Build–Test–Learn workflow for enzyme discovery from metagenomes, in which protein language model embeddings select candidates and cell-free protein synthesis on a biofoundry measures what they do.

- Narrowed ~5.3M ORFs from 16 environmental metagenomes to a tractable candidate set, recovering sequences too distant for homology search to return, by generating ProtT5 embeddings as a 1,024-dimensional vector database and ranking by cosine similarity to the mean embedding of 98 structurally characterized bacterial lipases.
- Raised cell-free expression yield ~2.5× over a commercial kit by GFP comparison, and detected pNPO/pNPB/pNPP activity in the CalB control that the kit had not, by preparing S30 extract in-house from a GroEL/GroES-overexpressing strain; shortened library construction by carrying the T7 promoter/terminator in the primer 5′ tails, so that rounds run under identical conditions.
- Identified 43 hit candidates (5.0%), 16 of them active on two or more substrates, by screening 855 candidate ORFs from 9 libraries against 3 pNP substrates on a 96/384-well automated workflow (Echo 525 acoustic dispenser, automated liquid handlers) and replacing ad-hoc thresholds with a robust Z-score criterion ($Z \geq 3$ plus positive-control comparison).
- Making the selection criterion itself update each round, by training an activity prediction model on the Round 1 measurements (frozen ESM-C encoder with trainable pooling and projection heads) in which the ORFs with no detected activity serve as experimentally demonstrated negatives rather than negatives guessed from sequence distance.
- Structural interpretation of hits with ESMFold and Foldseek fold-similarity search, plus FoldDisco analysis of the discontinuous catalytic motif (Ser–His–Asp triad, oxyanion hole).
- Digitized the lab notebook and built a reagent, instrument and sample database so that experimental records are consistent enough to be reused as data.

### Undergraduate Researcher — Research Center for Plant Plasticity, Seoul National University

Dec 2021 – Feb 2022

Temperature-dependent protein changes in *Arabidopsis*; yeast two-hybrid, insert cloning, restriction digestion. Wrote a lab-specific protocol still used as training material.

### Undergraduate Researcher — Infection and Inflammatory Disease Lab, Sungkyunkwan University

Sep 2021 – Dec 2021

Immune-cell changes in diet-induced obese C57BL/6 mice; cell culture, ELISA, western blot, flox genotyping.

## INDUSTRY EXPERIENCE

### Quality Assurance, Biologics — Hanmi Pharmaceutical, Pyeongtaek Bio Plant

Jun 2023 – Feb 2025 (1 yr 8 mo), full-time

Biologics quality assurance group at a commercial GMP plant producing pre-filled-syringe biologics, covering regulatory inspections, deviations and change control, and supplier quality.

- Took the record at MFDS and FDA inspections as the English scribe, translating the inspectors' questions and the answers into the Korean record in real time, and prepared the documentation and evidence they asked for.
- Owned the supply-chain side of those inspections: prepared the supporting evidence for supplier and supply-chain lines of questioning and answered them during the audit.
- Reviewed manufacturing records and quality documentation under GMP; preparing for an inspection meant tracing a product's history through its records and resolving every point where two documents disagreed.
- Investigated manufacturing deviations, identified root causes and managed the associated change control.
- Managed supplier qualification and supply-chain quality documentation, and supported overseas partner audits and meetings as interpreter.

## SELECTED PROJECTS

### MetaForge — Deep Learning-Based Discovery of Moonlighting Enzymes

IDEA-B Synthetic Biology Competition, MSIT / Korea Synthetic Biology Council · Mar – Oct 2026 · student team of 4 (with an advising researcher and professor) · my role: candidate selection and validation design

**Outcome.** Passed first-round review and was funded as a commissioned research project at KRIBB. Completed ~10,000 screening conditions — 380 metagenomic candidates × 7 substrates × 4 reaction conditions — in three days, inside an eight-month budget.

**How.** Built the case for the premise first: a literature and database review showed that multifunctional proteins are over-represented among proteins linked to drug side effects, which is what makes the single-function assumption behind conventional enzyme discovery worth attacking. Then cut the scope to fit the budget — 14 enzyme–substrate reactions collected from BRENDA, UniProt and prior literature were reduced to 7 validated systems by pre-screening controls that dropped every combination where the biosensor cross-detected the substrate or lacked sensitivity. Discarding half the candidates early is what made the screen finishable at all.

### K-Software Empowerment BootCamp — 640 hours of software training

SW Talent Development Program (IITP) at Sungkyunkwan University · Oct 2022 – Feb 2023 · Excellence Award, healthcare track

A 640-hour intensive in data processing, artificial intelligence and backend development, taken alongside a biology degree. Four individual AI/data projects, including an oral medication image classifier and an analysis of COVID-19 case data.

**MAKEAT**, the team project, is an app that logs meals by classifying photographs of food; it won the award in the healthcare track. I managed the database and volunteered for the Android/Flutter development role when no one else would take it, learning the stack during the project. This is where the computational half of my current research started.

### LLM agents for research and workflow automation

Personal projects, 2025 – present · repositories private, walkthrough available on request

Built agents for the repetitive parts of research: paper translation and summary with a fixed glossary, meeting minutes from recordings, a lab notebook assistant that expands a short plan into a templated record, and a job-posting digest that collects, filters and mails a daily report unattended in CI. Each is a Claude Code project with its own skills, commands and hooks.

## PUBLICATIONS & PRESENTATIONS

**Global Representations in Overlooked Single-token Embeddings of Protein Language Models.** S. Heo, **Y. Kang**, G. Park, S.-G. Lee, D.-H. Lee, H. Kim. Poster Q-12, 52nd Annual Meeting of the Korean Society for Microbiology and Biotechnology (KMB 2025), Busan, Jun 2025. — Second of six authors; single-token embeddings improved bacterial essential-gene prediction by up to 17% over average pooling.

## DIFFERENTIATING STRENGTHS

- **I trace errors to their cause.** Deviation investigation and change control in a GMP plant, and in research, replacing hand-set thresholds with a criterion that can be stated, repeated and audited.
- **I remove wasted steps.** Library construction cut from four steps to two, a commercial kit replaced with in-house extract at ~2.5× the yield, repetitive work moved onto automated liquid handlers and into LLM agents.
- **I work on both sides of the evidence.** I build the models that generate a result and run the assays that test it, with the documentation habits of a regulated environment — the question regulators are now asking of AI-derived evidence.

## SKILLS

**Computational**

Python (PyTorch), R, Linux/GPU server workflows; protein language models (ProtT5, ESM-C, ESM-2) and embedding database construction (HDF5); ESMFold, AlphaFold3, RFdiffusion, LigandMPNN; Foldseek, FoldDisco; UMAP and contrastive representation learning; NGS/multi-omics analysis.

**Experimental**

Cell-free protein synthesis and S30 extract preparation; primer design, PCR, Gibson assembly, transformation; enzyme activity assays (pNP substrates), SDS-PAGE; cell culture, ELISA, western blot, yeast two-hybrid.

**Automation & instruments**

Echo 525 acoustic dispenser, Zephyr G3 liquid handler, Trobot II automated PCR, 5400 Fragment Analyzer, microplate readers (Victor Nivo, Tecan Spark).

**Quality & regulatory**

GMP; regulatory inspection support (MFDS, FDA); deviation investigation and root cause analysis; change control; supplier and supply-chain quality management; technical documentation in Korean and English.

**Languages**

Korean (native); English (OPIc IH, 2026; TOEIC 960, 2021 — certificate expired; elementary schooling in San Jose, California; regulatory inspection interpreting experience).

## AWARDS & ACTIVITIES

Magna Cum Laude, Sungkyunkwan University (Feb 2023) · Academic Excellence Scholarship, highest tier, Sungkyunkwan University (2022) · Excellence Award, Healthcare Track, K-Software Empowerment BootCamp (Feb 2023) · KAIST Language Buddy Program, participant (Sep 2025 – Aug 2026), all team coordination in English · WISET Junior Mentor, STEM career mentoring for high school students (2017)